1. Display department name and manager first name.

```
Select  b.first_name, d.department_name
From employees a
Join employees b on a.manager_id = b.employee_id
Join departments d on b.department_id =d.department_id
Group by b.first_name, d.department_name;
```

```
d5_96305_sanchit> Select  b.first_name, d.department_name
    -> From employees a
    -> Join employees b on a.manager_id = b.employee_id
    -> Join departments d on b.department_id =d.department_id
    -> Group by b.first_name, d.department_name;
+------------+-----------------+
| first_name | department_name |
+------------+-----------------+
| Steven     | Executive       |
| Neena      | Executive       |
| Lex        | Executive       |
| Alexander  | IT              |
| Nancy      | Finance         |
| Den        | Purchasing      |
| Matthew    | Shipping        |
| Adam       | Shipping        |
| Payam      | Shipping        |
| Shanta     | Shipping        |
| Kevin      | Shipping        |
| John       | Sales           |
| Karen      | Sales           |
| Alberto    | Sales           |
| Gerald     | Sales           |
| Eleni      | Sales           |
| Michael    | Marketing       |
| Shelley    | Accounting      |
+------------+-----------------+
18 rows in set (0.058 sec)

d5_96305_sanchit>
```

2. Display department name, manager name, and city

```
Select  b.first_name, d.department_name, l.city
From employees a
Join employees b on a.manager_id = b.employee_id
Join departments d on b.department_id =d.department_id
Join locations l on d.location_id = l.location_id
Group by b.first_name, d.department_name, l.city;
```

```
d5_96305_sanchit> Select  b.first_name, d.department_name, l.city
    -> From employees a
    -> Join employees b on a.manager_id = b.employee_id
    -> Join departments d on b.department_id =d.department_id
    -> Join locations l on d.location_id = l.location_id
    -> Group by b.first_name, d.department_name, l.city;
+------------+-----------------+---------------------+
| first_name | department_name | city                |
+------------+-----------------+---------------------+
| Steven     | Executive       | Seattle             |
| Neena      | Executive       | Seattle             |
| Lex        | Executive       | Seattle             |
| Alexander  | IT              | Southlake           |
| Nancy      | Finance         | Seattle             |
| Den        | Purchasing      | Seattle             |
| Matthew    | Shipping        | South San Francisco |
| Adam       | Shipping        | South San Francisco |
| Payam      | Shipping        | South San Francisco |
| Shanta     | Shipping        | South San Francisco |
| Kevin      | Shipping        | South San Francisco |
| John       | Sales           | Oxford              |
| Karen      | Sales           | Oxford              |
| Alberto    | Sales           | Oxford              |
| Gerald     | Sales           | Oxford              |
| Eleni      | Sales           | Oxford              |
| Michael    | Marketing       | Toronto             |
| Shelley    | Accounting      | Seattle             |
+------------+-----------------+---------------------+
18 rows in set (0.013 sec)
```

3. Display country name, city, and department name.

   Select d.department_name , l.city, c.country_name

   From departments d

   Join locations l on l.location_id = d.location_id

   Join countries c on c.country_id = l.country_id;

```
d5_96305_sanchit> Select d.department_name , l.city, c.country_name
    -> From departments d
    -> Join locations l on l.location_id = d.location_id
    -> Join countries c on c.country_id = l.country_id;
+---------------------+---------------------+----------------------------------------+
| department_name     | city                | country_name                           |
+---------------------+---------------------+----------------------------------------+
| Administration      | Seattle             | United States of America               |
| Marketing           | Toronto             | Canada                                 |
| Purchasing          | Seattle             | United States of America               |
| Human Resources     | London              | United Kingdom of Great Britain and Nort |
| Shipping            | South San Francisco | United States of America               |
| IT                  | Southlake           | United States of America               |
| Public Relations    | Munich              | Germany                                |
| Sales               | Oxford              | United Kingdom of Great Britain and Nort |
```

4. Display job title, department name, employee last name, starting date for all jobs from 2011 to 2015.

   Select j.job_title,d.department_name, e.last_name,h.start_date
   From employees e
   Join departments d on e.department_id = d.department_id
   Join jobs j on e.job_id = j.job_id
   Join job_history h on j.job_id = h.job_id
   Where year(h.start_date) between 2011 and 2015.

```
d5_96305_sanchit> Select j.job_title,d.department_name, e.last_name,h.start_date
    -> From employees e
    -> Join departments d on e.department_id = d.department_id
    -> Join jobs j on e.job_id = j.job_id
    -> Join job_history h on j.job_id = h.job_id
    -> Where year(h.start_date) between 2011 and 2015;
+--------------------------+-----------------+-----------+------------+
| job_title                | department_name | last_name | start_date |
+--------------------------+-----------------+-----------+------------+
| Public Accountant        | Accounting      | Gietz     | 2012-07-01 |
| Accounting Manager       | Accounting      | Higgins   | 2011-10-28 |
| Programmer               | IT              | James     | 2011-01-13 |
| Programmer               | IT              | Miller    | 2011-01-13 |
| Programmer               | IT              | Williams  | 2011-01-13 |
| Programmer               | IT              | Jackson   | 2011-01-13 |
| Programmer               | IT              | Nguyen    | 2011-01-13 |
| Marketing Representative | Marketing       | Davis     | 2014-02-17 |
+--------------------------+-----------------+-----------+------------+
8 rows in set (0.006 sec)
```

5. . Display job title and average salary of employees.

   Select j.job_title, avg(e.salary)
   From employees e
   Join jobs j on e.job_id = j.job_id
   Group by j.job_title;

```
d5_96305_sanchit> Select j.job_title, avg(e.salary)
    -> From employees e
    -> Join jobs j on e.job_id = j.job_id
    -> Group by j.job_title;
+---------------------------------+---------------+
| job_title                       | avg(e.salary) |
+---------------------------------+---------------+
| Public Accountant               |   8300.000000 |
| Accounting Manager              |  12008.000000 |
| Administration Assistant        |   4400.000000 |
| President                       |  24000.000000 |
| Administration Vice President   |  17000.000000 |
| Accountant                      |   7920.000000 |
| Finance Manager                 |  12008.000000 |
| Human Resources Representative  |   6500.000000 |
| Programmer                      |   5760.000000 |
| Marketing Manager               |  13000.000000 |
| Marketing Representative        |   6000.000000 |
| Public Relations Representative |  10000.000000 |
| Purchasing Clerk                |   2780.000000 |
| Purchasing Manager              |  11000.000000 |
| Sales Manager                   |  12200.000000 |
| Sales Representative            |   8350.000000 |
| Shipping Clerk                  |   3215.000000 |
| Stock Clerk                     |   2785.000000 |
| Stock Manager                   |   7280.000000 |
+---------------------------------+---------------+
19 rows in set (0.165 sec)

d5_96305_sanchit>
```

6. Display job title, employee name, and the difference between maximum salary for that job and salary of the employee.
   Select j.job_title, concat(e.first_name,” “,e.last_name) as full_name, (j.max_salary – e.salary) as diff
   From employees e
   Left outer join jobs j on e.job_id = j.job_id;

```
d5_96305_sanchit> select j.job_title, concat(e.first_name,' ',e.last_na
 jobs j on e.job_id = j.job_id;
+--------------------------------+-------------------+----------+
| job_title                      | full_name         | diff     |
+--------------------------------+-------------------+----------+
| Public Accountant              | William Gietz     |   700.00 |
| Accounting Manager             | Shelley Higgins   |  3992.00 |
| Administration Assistant       | Jennifer Whalen   |  1600.00 |
| President                      | Steven King       | 16000.00 |
| Administration Vice President  | Neena Yang        | 13000.00 |
| Administration Vice President  | Lex Garcia        | 13000.00 |
| Accountant                     | Daniel Faviet     |     0.00 |
| Accountant                     | John Chen         |   800.00 |
| Accountant                     | Ismael Sciarra    |  1300.00 |
| Accountant                     | Jose Manuel Urman |  1200.00 |
| Accountant                     | Luis Popp         |  2100.00 |
| Finance Manager                | Nancy Gruenberg   |  3992.00 |
| Human Resources Representative | Susan Jacobs      |  2500.00 |
| Programmer                     | Alexander James   |  1000.00 |
| Programmer                     | Bruce Miller      |  4000.00 |
```

7. Display the last name and job title of employees who earn a commission but do not belong to any department.

   Select e.last_name, j.job_title
   From employees e
   Join jobs j
   On e.job_id = j.job_id
   Where e.department_id is null;

```
d5_96305_sanchit> Select e.last_name, j.job_title
    -> From employees e
    -> Join jobs j
    -> On e.job_id = j.job_id
    -> Where e.department_id is null;
+-----------+----------------------+
| last_name | job_title            |
+-----------+----------------------+
| Grant     | Sales Representative |
+-----------+----------------------+
1 row in set (0.008 sec)
```

8. Display details of jobs that were done by any employee who is currently drawing more than 15000 of salary.

   Select j.job_id, j.job_title
   From employees e
   Join jobs j on e.job_id = j.job_id
   Where e.salary > 15000

```
d5_96305_sanchit> Select j.job_id, j.job_title
    -> From employees e
    -> Join jobs j on e.job_id = j.job_id
    -> Where e.salary > 15000
    -> ;
+---------+-------------------------------+
| job_id  | job_title                     |
+---------+-------------------------------+
| AD_PRES | President                     |
| AD_VP   | Administration Vice President |
| AD_VP   | Administration Vice President |
+---------+-------------------------------+
3 rows in set (0.011 sec)

d5_96305_sanchit>
```

9. Display department name, manager name, and salary of the managers, who have the experience of more than 12 years.
   Select d.department_name,a.manager_id, b.first_name as manager_name,b.salary, TIMESTAMPDIFF(YEAR, b.hire_date, NOW())
   From employees a left join employees b on a.manager_id = b.employee_id
   Join departments d on d.department_id = b.department_id
   Where a.manager_id is not null and TIMESTAMPDIFF(YEAR, b.hire_date, NOW()) >12;

```
d5_96305_sanchit> Select d.department_name,a.manager_id, b.first_name as manager_name,b.salary, TIMESTAMPDIFF(YEAR, b.hire_date, NOW())
    -> From employees a left join employees b on a.manager_id = b.employee_id
    -> Join departments d on d.department_id = b.department_id
    -> Where a.manager_id is not null and TIMESTAMPDIFF(YEAR, b.hire_date, NOW()) >12;
+----------------+------------+--------------+----------+--------------------------------------+
| department_name | manager_id | manager_name | salary   | TIMESTAMPDIFF(YEAR, b.hire_date, NOW()) |
+----------------+------------+--------------+----------+--------------------------------------+
| Executive      |        102 | Lex          | 17000.00 |                                   15 |
| Finance        |        108 | Nancy        | 12008.00 |                                   13 |
| Finance        |        108 | Nancy        | 12008.00 |                                   13 |
| Finance        |        108 | Nancy        | 12008.00 |                                   13 |
| Finance        |        108 | Nancy        | 12008.00 |                                   13 |
| Finance        |        108 | Nancy        | 12008.00 |                                   13 |
| Purchasing     |        114 | Den          | 11000.00 |                                   13 |
| Purchasing     |        114 | Den          | 11000.00 |                                   13 |
| Purchasing     |        114 | Den          | 11000.00 |                                   13 |
| Purchasing     |        114 | Den          | 11000.00 |                                   13 |
| Purchasing     |        114 | Den          | 11000.00 |                                   13 |
| Accounting     |        205 | Shelley      | 12008.00 |                                   13 |
+----------------+------------+--------------+----------+--------------------------------------+
12 rows in set (0.008 sec)
```

10. . Display employee name if the employee joined before his manager.
    select
    from employees a
    left join employees b on a.

11.

12. Display employee name and country in which he is working

Select e.first_name , c.country_name

From employees e join departments d on e.department_id = d.department_id

Join locations l on d.location_id = l.location_id

Join countries c on l.country_id = c.country_id

```
d5_96305_sanchit> Select e.first_name , c.country_name
    -> From employees e join departments d on e.department_id = d.department_id
    -> Join locations l on d.location_id = l.location_id
    -> Join countries c on l.country_id = c.country_id
    -> ;
+-------------+----------------------------------------+
| first_name  | country_name                           |
+-------------+----------------------------------------+
| Alexander   | United States of America               |
| Bruce       | United States of America               |
| David       | United States of America               |
| Valli       | United States of America               |
| Diana       | United States of America               |
| Matthew     | United States of America               |
| Adam        | United States of America               |
| Payam       | United States of America               |
| Shanta      | United States of America               |
| Kevin       | United States of America               |
| Julia       | United States of America               |
| Irene       | United States of America               |
| James       | United States of America               |
| Steven      | United States of America               |
| Laura       | United States of America               |
```

13. Display department name, average salary and number of employees with commission within the department.

Select d.department_name,avg(e.salary),count(*)

From employees e

Join departments d on e.department_id = d.department_id

Where e.commission_pct is not null

Group by d.department_name;

```
d5_96305_sanchit> Select        d.department_name,avg(e.salary),count(*)
    -> From employees e
    -> Join departments d on e.department_id = d.department_id
    -> Where e.commission_pct is not null
    -> Group by d.department_name;
+-----------------+---------------+----------+
| department_name | avg(e.salary) | count(*) |
+-----------------+---------------+----------+
| Sales           |   8955.882353 |       34 |
+-----------------+---------------+----------+
1 row in set (0.009 sec)

d5_96305_sanchit>
```

14. Display the count of employees, month and the year in which 3 or more employees joined in any department located in 'Seattle'.

Select count(e.employee_id), month(e.hire_date),year(e.hire_date)

From departments d join locations l on d.location_id = l.location_id

Join employees e on e.department_id = d.department_id

Where l.city = 'seattle'

group by month(e.hire_date), year(e.hire_date)

having count(e.employee_id)>2;

select

from employees e join departments d on e.department_id =

```
d5_96305_sanchit> Select count(e.employee_id), month(e.hire_date),year(e.hire_date)
    ->     From departments d join locations l on d.location_id = l.location_id
    ->      Join employees e on e.department_id = d.department_id
    ->      Where l.city = 'seattle'
    ->     group by month(e.hire_date), year(e.hire_date)
    -> having count(e.employee_id)>2;
+----------------------+--------------------+-------------------+
| count(e.employee_id) | month(e.hire_date) | year(e.hire_date) |
+----------------------+--------------------+-------------------+
|                    3 |                  9 |              2015 |
+----------------------+--------------------+-------------------+
1 row in set (0.148 sec)
```

15. Display employee name, job title, start date, and end date of past jobs of all employees with commission percentage null.

Select e.first_name, j.job_title, h.start_date, h.end_date, e.commission_pct

From employees e join jobs j on e.job_id = j.job_id

Join job_history h on e.employee_id = h.employee_id

Where e.commission_pct is null;

```
+------------+------------------------------+------------+------------+----------------+
| first_name | job_title                    | start_date | end_date   | commission_pct |
+------------+------------------------------+------------+------------+----------------+
| Neena      | Administration Vice President | 2007-09-21 | 2011-10-27 |           NULL |
| Neena      | Administration Vice President | 2011-10-28 | 2015-03-15 |           NULL |
| Lex        | Administration Vice President | 2011-01-13 | 2016-07-24 |           NULL |
| Den        | Purchasing Manager           | 2016-03-24 | 2017-12-31 |           NULL |
| Payam      | Stock Manager                | 2017-01-01 | 2017-12-31 |           NULL |
| Jennifer   | Administration Assistant     | 2005-09-17 | 2011-06-17 |           NULL |
| Jennifer   | Administration Assistant     | 2012-07-01 | 2016-12-31 |           NULL |
| Michael    | Marketing Manager            | 2014-02-17 | 2017-12-19 |           NULL |
+------------+------------------------------+------------+------------+----------------+
8 rows in set (0.141 sec)

d5_96305_sanchit>
```